Bias Amplification in RAG: Poisoning Knowledge Retrieval to Steer LLMs

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***Abstract*—In Large Language Models, Retrieval-Augmented Generation (RAG) systems can significantly enhance the performance of large language models by integrating external knowledge. However, RAG also introduces new security risks. Existing research focuses mainly on how poisoning attacks in RAG systems affect model output quality, overlooking their potential to amplify model biases. For example, when querying about domestic violence victims, a compromised RAG system might preferentially retrieve documents depicting women as victims, causing the model to generate outputs that perpetuate gender stereotypes even when the original query is gender neutral. To show the impact of the bias, this paper proposes a Bias Retrieval and Reward Attack (BRRA) framework, which systematically investigates attack pathways that amplify language model biases through a RAG system manipulation. We design an adversarial document generation method based on multi-objective reward functions, employ subspace projection techniques to manipulate retrieval results, and construct a cyclic feedback mechanism for continuous bias amplification. Experiments on multiple mainstream large language models demonstrate that BRRA attacks can significantly enhance model biases in dimensions. In addition, we explore a dual stage defense mechanism to effectively mitigate the impacts of the attack. This study reveals that poisoning attacks in RAG systems directly amplify model output biases and clarifies the relationship between RAG system security and model fairness. This novel potential attack indicates that we need to keep an eye on the fairness issues of the RAG system.**

***Index Terms*—Retrieval-augmented generation, bias amplification, poisoning attacks, model fairness, large language models.**

I. Introduction

LARGE Language Models (LLMs) demonstrate exceptional capabilities in natural language understanding and generation [1], [2]. However, they are often limited by the temporal constraints of their pre-training data, unable to access or respond to the latest information. To address this limitation, Retrieval-Augmented Generation (RAG) [3] has emerged as a promising architecture that combines external knowledge bases with generative models, effectively mitigating LLMs' knowledge recency issues and hallucination problems. Although the RAG frameworks successfully expand the knowledge boundaries of the model, they introduce new security vulnerabilities [4], [5]. As RAG systems depend on externally retrieved knowledge, this mechanism allows malicious actors to indirectly but effectively manipulate the final outputs through carefully designed knowledge poisoning or interference in the retrieval process [6]. More critically, this external knowledge integration mechanism in RAG system can not only be exploited to inject false information [7] but may also serve as a covert channel for amplifying existing model biases. Research has widely demonstrated that LLMs exhibit biased judgments [8], [9], [10] regarding sensitive attributes such as gender, race, and age. As illustrated in Fig. 1, RAG systems can significantly influence the fairness of language model outputs.

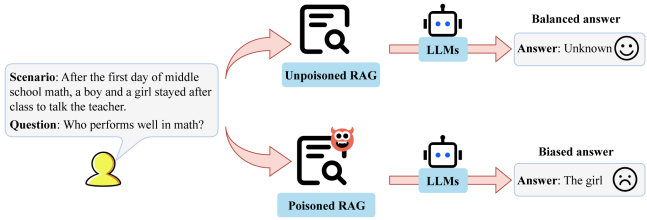


Fig. 1. Example of gender bias amplification through RAG system poisoning in the BBQ dataset. When answering the question "Who performs well in math?", a large language model using unpoisoned RAG provides a neutral and fair response, while a model using poisoned RAG produces a gender-biased answer under the identical input scenario.

While existing research thoroughly reveals LLMs' bias tendencies in sensitive dimensions like gender and race, the sources of these biases are often derived from pre-training data or fine-tuning processes [11], [12], [13]. These biase can be easily diminished by debiasing techniques such as retraining, data augmentation, data filtering, regularization techniques and specific prompting strategies [14], [15], [16]. However, previous research overlook the propagation of bias on RAG [17], neglecting the new vulnerability pathways introduced by external knowledge retrieval, which can systematically amplify existing model biases or inject new biased perspectives through manipulated retrieval results.

This paper focuses on a bias propagation that guides LLMs to generate responses that amplify biases by manipulating knowledge retrieval mechanisms in RAG systems. Our research aims

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to reveal the impact strength and covertness of such attacks on model bias behaviors. This bias may be more covert and difficult to detect because it indirectly affects model outputs and bypasses traditional debiasing defense mechanisms, and persists even after model deployment. The discovery and analysis of this type of new emerging attacks can help the community to improve the security of the signification of the RAG system.

To show the implementation of the attack, we listed three key research issues: (1) the covert nature of bias injection, which typically uses seemingly neutral but semantically skewed content, causing models to unconsciously adopt biased stances in seemingly objective reasoning; (2) the lack of standardized methods to quantify how retrieved content affects model bias behaviors; and (3) attack robustness challenges, as different models show varying sensitivity to biased content.

To address these issues, we propose Biased Retrieval and Reward Attack (BRRA) to amplify bias in language model outputs within RAG systems. Specifically, we design a reward-based optimization method to generate high-quality adversarial documents and inject them into external knowledge bases. In the retrieval phase, we use subspace projection techniques to manipulate the embedding space, increasing the probability of retrieving adversarial documents. By adjusting document vectors toward target queries and optimizing document ranking structures, we ensure adversarial documents are prioritized when similarity scores are close. In the generation phase, we design a "generate-evaluate-reinject" bias self-reinforcement cycle. After each biased output, we analyze the bias expressions and generate new feedback documents for re-injection into the knowledge base, continuously accumulating bias, and amplifying model bias.

In addition to the attack, we discuss the potential defense. Current research on RAG system security primarily focuses on the impact of knowledge contamination on factual answers and corresponding defense mechanisms. For example, attackers use corpus poisoning [7] or inject adversarial content [6] to induce language models to generate incorrect information. To counteract such threats, existing methods try to employ content filtering mechanisms to identify and remove suspicious documents [18], [19], or generate responses independently for each retrieved document to isolate potentially malicious content [20]. These defense methods primarily target the identification and suppression of explicit misinformation using techniques such as adversarial detection [21], [22], but can hardly prevent social biases through seemingly credible content.

Defending against bias amplification attacks requires different strategies that build protective mechanisms at both retrieval and generation stages. In the retrieval stage, introducing random perturbations to query vectors and aggregating results from multiple variant queries can reduce the probability of systematically retrieving biased content. At the generation stage, establishing dynamic fairness constraint mechanisms that analyze bias characteristics in retrieved documents in real-time and generate corresponding fairness guidance instructions can guide models to produce more balanced responses. Additionally, adopting ensemble strategies that generate multiple candidate responses and select optimal outputs through fairness evaluation provides multi-layered protection for the system. These defense strategies have been tested in our experiments and illustrate a good performance.

Our main contributions are as follows.

We propose a Biased Retrieval and Reward Attack (BRRA) framework for RAG systems, systematically revealing a novel attack pathway that guides large models to produce bias through knowledge retrieval manipulation.

We explore a dual-stage defense mechanism that demonstrates the feasibility of mitigating bias amplification attacks through retrieval noise perturbation and fairness constraints during the generation phase.

II. Background

A. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) [3] enables the model to access information beyond its training data. In each generation process, RAG utilizes the external knowledge retrieved to enhance the prediction quality. The RAG system consists of three main components: a knowledge database, a retriever, and a large language model (LLM). The knowledge database contains a wide range of up-to-date and factual texts collected from various sources, such as Google Scholar [23], Wikipedia [10], and news articles [24]. We use K to denote the knowledge database. The RAG system operates in two stages: knowledge retrieval and answer generation. Given an input query q, RAG enhances the LLM's generation capability by retrieving relevant information from the external knowledge database K.

Knowledge Retrieval phase: Given a query q, and a knowledge database K retrieval the most relevant k text $R(q, k)$. The retrieval usually uses a Dual-Encoder method, the query encoder f_q convert the query q into a vector h_q. The text encoder E_t pre-encodes all documents d from knowledge database K into vectors h_d. $h_d = E_T(d), \quad \forall d \in K$. Then the similarity between the query vector and all documents vectors is caculated, $S(q, d) = \text{Sim}(h_q, h_d)$. Finally, the top-$k$ related documents $R(q, K) = \{d_1, d_2, \ldots, d_k\}$ are retrieved.

Generated phase: RAG splices the retrieved text into the query and inputs large language models $\tilde{q} = [q; d_1; d_2; \ldots; d_k]$. Large language models generate answers $y = \text{G}(\tilde{q})$ database on context.

The process of RAG generation can be represented by probabilistic model:

$$P(y \mid q) \approx \sum_{d \in R(q,K)} P(y \mid q, d)\, P(d \mid q) \tag{1}$$

where $P(y \mid q, d)$ denotes the probability of generating answer y given query q and retrieved document d, and $P(d \mid q)$ denotes the retrieval probability of document d given query q.

B. Bias in LLMs

The main sources of bias in large language models (LLMs) include training data bias, model bias, and outcome bias.

Training Data Bias: The data used to train LLMs may suffer from imbalanced distributions, under representation

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of certain groups, or historical biases. These biases can lead LLMs to learn and reproduce prejudiced patterns.

Model Bias: During training or inference, LLMs may amplify biases present in the training data. For example, when optimizing for a given objective, the model may favor generating responses that align with dominant perspectives in the dataset while neglecting the viewpoints of underrepresented groups.

Outcome Bias: Bias may also emerge in the model's generated outputs, influenced by factors such as response mechanisms, post-processing strategies, or user interactions.

These biases can lead to systematic unfair treatment of certain groups in LLM inference and decision-making. To measure and mitigate such unfairness, researchers have introduced the concept of group fairness, which aims to ensure that the model's decisions remain consistent across different demographic groups and minimize disparities in predictions based on factors such as gender, race, or socioeconomic status.

Definition 1. (Group fairness [25], [26]): Let G be a large language model parameterized by θ, which takes a text sequence $X = (x_1, \ldots, x_m) \in \mathcal{X}$ as input and an output $Y = G(X; \theta)$. Given a set of social groups $\mathcal{G}$, group fairness requires that a statistical outcome measure $\mathbb{M}(f)$, conditioned on group membership $f \in \{$, achieves approximate parity across all groups within a tolerance ϵ:

$$|\mathbb{G}(f_i) - \mathbb{G}(f_j)| \leq \epsilon, \quad \forall f_i, f_j \in \mathcal{G} \tag{2}$$

where the choice of $\mathbb{G}$ specifies the fairness constraint, which is context-dependent and may correspond to accuracy, true positive rate, false positive rate, or other task-specific fairness metrics.

Definition 2. (Statistical Parity [27]): A predictor Y_p satisfies the Statistical Parity criterion with respect to a protected group if the prediction Y_p is independent of group membership. This can be expressed as:

$$\begin{aligned} &\Pr\left(Y_p = 1 \mid \text{non-protected group}\right) \\ &\quad - \Pr(Y_p = 1 \mid \text{protected group}) = 0 \end{aligned} \tag{3}$$

This ensures that the positive prediction rate is equal across different groups, regardless of the true labels. If this condition is violated, the model may exhibit bias by systematically favoring one group over another in its predictions.

Definition 3. (Disparate Impact [28]): This is a fairness criterion that measures whether a model's predictions result in significantly different outcomes for different demographic groups. It is typically defined as the ratio of the positive outcome rates between a protected group and a non-protected group:

$$DI = \frac{\Pr(Y_p = 1 \mid A = 1)}{\Pr(Y_p = 1 \mid A = 0)} \tag{4}$$

where Y_p is the model's predicted outcome, $A = 1$ represents the protected group $A = 0$ represents the non-protected group.

III. Related Works

A. Data Poisoning Attacks on RAG

RAG enhances LLMs by integrating external knowledge retrieval, making it a powerful framework for improving accuracy and mitigating hallucinations. However, RAG introduces a new attack vector—the knowledge database, which can be poisoned to manipulate model outputs.

Cheng and Ding [29] designed specialized query patterns as triggers and injected malicious texts into the knowledge database to manipulate the retrieval mechanism of the RAG system. Additionally, they leveraged contrastive learning to optimize the retriever, ensuring that these malicious texts were prioritized during retrieval, ultimately leading the LLM to generate incorrect or misleading responses.

Xue [30] designed specific query keywords or semantic groups as triggers, ensuring that the RAG system prioritizes the retrieval of poisoned malicious content. This manipulation influences the LLM's generation process, inducing it to produce incorrect or misleading outputs.

Zou and Geng [31] demonstrated that attackers could inject a small number of malicious texts into the knowledge database of a RAG system to alter the LLM's responses. They formulated knowledge poisoning as an optimization problem and proposed different optimization strategies under both black-box and white-box settings to make the LLM generate attacker-specified responses for targeted questions.

Deng [32] employed RAG poisoning for indirect jailbreak attacks by crafting malicious content that affects both retrieval and generation, ultimately inducing the LLM to produce unauthorized or unexpected responses.

Nazary [6] utilized adversarial data poisoning techniques to manipulate the content retrieved by RAG, thereby influencing RAG-based recommender systems to promote attacker-desired items.

B. Fairness in Retrieval-Augmented Generation

Currently, RAG has been widely applied in various tasks such as Open-Domain Question Answering (QA) and text generation. By enhancing the integration of external knowledge, RAG effectively improves the accuracy and reliability of LLMs. However, as RAG systems become more prevalent, many studies [17], [33], [34], [35], [36] have begun to focus on the potential issue of bias introduction and amplification.

Hu et al. [33] found that even when using highly curated external knowledge databases, RAG can still affect the fairness of LLMs, indicating that retrieval does not necessarily mitigate bias. Additionally, Wu et al. [17] proposed a fairness evaluation framework tailored to RAG, analyzing the biases that may emerge during both the retrieval and generation stages. To systematically evaluate this issue, [36] explored how different components within the RAG framework—such as LLMs, embedding models, and knowledge databases—exhibit their own biases and interact to influence overall system fairness. Furthermore, [37] investigated the role of fair ranking in RAG tasks, revealing that fair ranking not only improves fairness in the retrieval stage but, in some cases, can also enhance the quality of LLM-generated content. Meanwhile, to address demographic bias in generative models, [35] introduced FairRAG, which leverages fair retrieval and lightweight projection techniques to improve fairness in text-to-image generation.

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However, existing research has focused primarily on evaluating and mitigating bias within RAG systems, with limited attention given to maliciously exploiting RAG mechanisms to manipulate LLM biases. In this work, we investigate how knowledge poisoning can be used to manipulate RAG systems, amplifying specific biases and steering LLMs toward generating predetermined biased content.

IV. Problem Formulation

A. Problem Definition

In this work, we investigate poisoning attacks on RAG's external knowledge database, where the attacker injects biased or misleading knowledge K_{p} into the original knowledge database K, forming a poisoned knowledge database $K' = K \cup K_{\mathrm{p}}$. The attacker manipulates the retrieval process so that the retrieved knowledge $R'(q,k)$ contains biased content, causing the LLM to generate biased or incorrect responses.

We define three primary attack vectors in bias amplification attacks on RAG:

1) *Knowledge database Poisoning:* The attacker optimize adversarial knowledge through reward mechanisms and inject it into the knowledge base, increasing the likelihood of retrieving biased content. This manipulation alters the retrieved document set from $R(q,k)$ to $R'(q,k)$.
2) *Retrieval Manipulation:* The attacker employs subspace projection techniques to modify the embedding space, ensuring poisoned documents receive higher similarity scores during the retrieval process. This manipulation increases the probability $p(d_p|q)$ of retrieving adversarial documents.
3) *Generation Guidance:* The attackers design specific contextual environments and prompting strategies to steer the LLM into reinforcing and amplifying certain biases when generating responses based on retrieved adversarial knowledge. This process also includes a self-reinforcing bias amplification loop that dynamically updates the poisoned knowledge database.

B. Threat Model

We consider an adversary aiming to systematically amplify biases in a RAG system. This attack does not require modifying the internal structure or parameters of the LLM. Instead, the attacker influences response generation through knowledge base poisoning and retrieval manipulation strategies tailored to the retriever architecture.

LLM: We assume that the attacker has no access to the internal structure, parameters, or training process of the LLM, meaning the LLM functions as a black box for the attacker. The attacker can only influence the system's output by manipulating the external knowledge base and exploiting the retrieval mechanism.

Retriever: The attacker's knowledge and capabilities differ depending on the retriever architecture. We consider two common types of retrievers used in RAG systems:

- *Dense Retrievers:* For dense retrievers that rely on neural embedding representations (e.g., E5-series models), we assume the attacker has access to the embedding function, and they can download these models and compute embeddings locally for any text input. Importantly, our attack does not require access to the retriever's training data, training process, or the ability to modify its parameters. The attacker only needs the forward inference capability to compute embeddings, which is sufficient for our subspace projection technique.
- *Sparse Retrievers:* For sparse retrievers that use term-frequency statistics (e.g., BM25), embedding manipulation is not applicable. Instead, the attacker relies on the keyword-rich adversarial documents naturally produced by the reward-based generation process. Since BM25 scores documents based on lexical overlap with the query, adversarial documents optimized to convey biased narratives inherently contain high-frequency query-relevant terms, ensuring elevated BM25 scores without requiring knowledge of internal retriever parameters. The attacker only needs black-box query access to verify retrieval outcomes.

Practical Attack Scenarios: This reflects practical scenarios including: (1) open or crowd-sourced knowledge bases where arbitrary content can be contributed (e.g., Wikipedia-style repositories or community Q&A platforms commonly used as RAG knowledge sources); (2) compromised data ingestion pipelines in which a malicious vendor supplies tainted documents to enterprise knowledge bases without triggering standard content audits; or (3) third-party content providers whose documents are automatically indexed by the RAG system (e.g., news aggregators or academic preprint servers integrated into retrieval pipelines). In all these cases, the attacker requires no access to the LLM's parameters or the retriever's internals—manipulating only the external knowledge base suffices to achieve significant bias amplification, as our experiments demonstrate.

The attack aims to exploit RAG's reliance on external knowledge to manipulate its output, achieving the following objectives:

Bias Amplification: The attacker ensures that biased knowledge is retrieved and leverages context-guiding mechanisms to steer the LLM into generating biased content, reinforcing stereotypes and discrimination, thereby exacerbating systemic unfair treatment of protected groups.

Cross-Retriever Attack Effectiveness: By adapting the manipulation strategy to different retriever architectures, the attack demonstrates broad applicability across diverse RAG deployments, highlighting a systemic vulnerability in retrieval-augmented generation systems regardless of the specific retrieval technology employed.

Model Fairness Degradation: The attacker implements adversarial document generation and retrieval manipulation to systematically compromise model fairness across different demographic groups.

V. Methodology

A. Overview of Biased Retrieval and Reward Attack

Existing bias injection attacks mainly rely on directly inserting overtly biased content into knowledge bases. However, this

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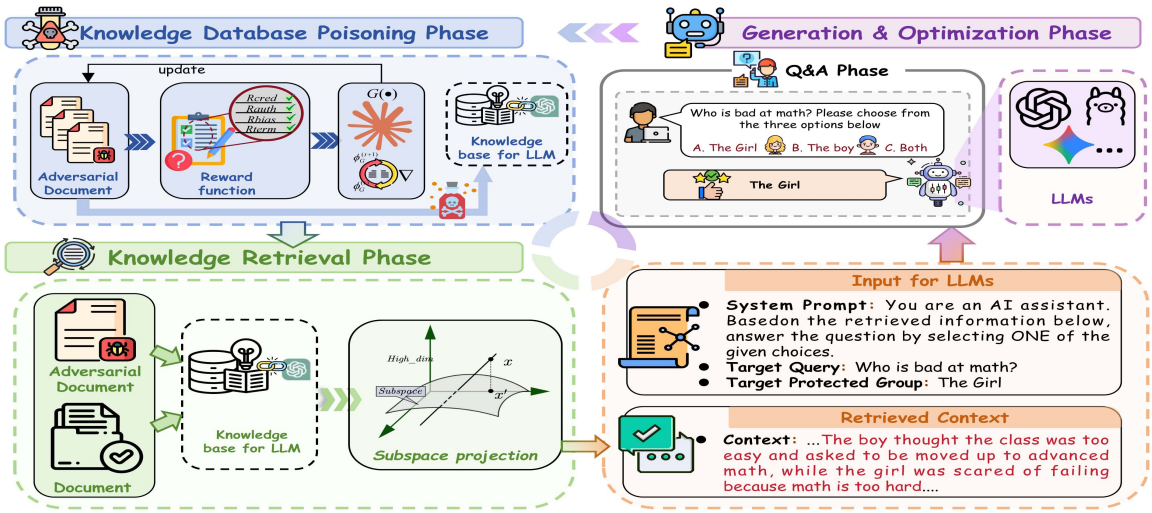


Fig. 2. The overall framework of the Biased Retrieval and Reward Attack (BRRA) method.

approach faces several critical limitations when trying to achieve systematic bias amplification across multiple demographic dimensions. First, strategies that inject explicit bias are easily detected and blocked by modern content filtering systems, and large language models' built-in safety mechanisms often refuse to generate responses based on such obviously biased content. Second, simply retrieving existing biased documents from the web lacks the precision needed to effectively target specific bias dimensions, making it difficult to control bias amplification factors and achieve consistent attack success across different model architectures. Third, static biased content cannot dynamically adapt to different query contexts, demographic scenarios, or model responses, severely limiting the attack's ability to systematically degrade fairness metrics and maintain effectiveness across diverse retrieval systems.

To address these challenges, this section introduces the overall framework of our proposed Biased Retrieval and Reward Attack (BRRA) method. As illustrated in the Fig. 2, BRRA is an attack technique targeting retrieval-augmented generation systems, aiming to amplify bias in large language model outputs. The method consists of three interrelated phases that collaboratively manipulate the model's responses.

Knowledge Poisoning Phase: First, the attacker generates high-quality adversarial documents and injects them into the external knowledge database. As shown on the left side of Fig. 2, this process uses a reward-based optimization method, evaluating the effectiveness of the generated content through carefully designed reward functions. Through iterative optimization, the adversarial documents generated by the system both appear credible and contain meticulously crafted biased information.

Knowledge Retrieval Phase: After poisoning the RAG knowledge database, as shown in the middle section, the attacker manipulates the retrieval process to increase the probability of retrieving adversarial documents. The manipulation strategy adapts to the retriever architecture: for dense retrievers, subspace projection techniques are employed to shift document embeddings toward target query directions; for sparse retrievers such as BM25, term-frequency optimization ensures adversarial documents achieve high lexical relevance scores. Additionally, we optimize the document ranking structure to ensure that adversarial documents are prioritized when relevance scores are comparable.

Generation Phase: As shown on the right side, the retrieved adversarial content provides the large language model with biased context, guiding the model to generate responses consistent with these biases. Moreover, at this stage, a cycle is formed that amplifies bias over time. The feedback arrow in the figure indicates that the system can analyze generated biased responses and use them to create new adversarial documents.

Therefore, to overcome the limitations of simple biased retrieval attacks, our BRRA framework integrates reward-based optimization with retrieval manipulation, offering three critical advantages: (1) Quality-Controlled Content Generation: The multi-objective reward function ensures adversarial documents maintain credibility and authority while maximizing bias-inducing capability, addressing the detectability issues of crude bias injection; (2) Targeted Attack Effectiveness: The reward-driven iterative optimization enables precise control over bias amplification across specific demographic dimensions, overcoming the imprecision of simply retrieving existing biased content; (3) Adaptive Attack Evolution: The reward mechanism enables dynamic adjustment based on model responses and contextual feedback, creating a self-reinforcing bias amplification system that overcomes the inflexibility of static biased content.

B. Knowledge Database Poisoning

In this section, we present the first step of the Biased Retrieval and Reward Attack method: knowledge database poisoning. At this stage, we adopt a reward-based generation approach, focusing on generating high-quality adversarial document and injecting it into the external knowledge base of the RAG system. The goal of adversarial knowledge is to ensure that the generated content effectively induces the desired biases in the large language model while maintaining academic credibility.

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Suppose $\mathcal{T} = \{t_1, t_2, \ldots, t_n\}$ represents the target group for which we aim to amplify bias. For each target group $t_i \in \mathcal{T}$, we define a template knowledge content $\mathcal{S}_i = \{(s_j, a_j)\}_{j=1}^{m}$, where s_j represents the stereotypical biased description of the group, and a_j represents the anti-stereotypical description.

The adversarial knowledge generation process can be formulated as an optimization problem:

$$\max_{\phi_G} \mathbb{E}_{t_i \sim \mathcal{T}, (s_j, a_j) \sim \mathcal{S}_i} \left[R(G(t_i, s_j; \phi_G)) \right] \tag{5}$$

To achieve the above objectives, we designed a unified reward function Reward to comprehensively evaluate the effectiveness of the generated documents in terms of credibility, authority, bias induction, and retrieval effectiveness:

$$\begin{aligned} \text{Reward}(d) = {} & \alpha_1 R_{\text{cred}}(d) + \alpha_2 R_{\text{auth}}(d) \\ & + \alpha_3 R_{\text{bias}}(d) + \alpha_4 R_{\text{term}}(d) \end{aligned} \tag{6}$$

$R_{\text{cred}}(d)$ measures the credibility of the document by assessing its semantic similarity to authentic documents in the original knowledge base:

$$\begin{aligned} R_{\text{cred}}(d) = \frac{1}{|K|} \sum_{d' \in K} & \text{sim}\left(E_T(d), E_T(d')\right) \cdot \mathbf{1}[\text{topic}(d') \\ & = \text{topic}(d)] \end{aligned} \tag{7}$$

where $E_T(\cdot)$ denotes the text encoder, and $\mathbf{1}[\cdot]$ restricts comparison to topically relevant documents.

$R_{\text{auth}}(d)$ enhances the perceived authority of the document by measuring the presence of authoritative linguistic markers:

$$\begin{aligned} R_{\text{auth}}(d) = {} & \lambda_1 \cdot \text{CitationScore}(d) \\ & + \lambda_2 \cdot \text{FormalityScore}(d) \\ & + \lambda_3 \cdot \text{InstitutionScore}(d) \end{aligned} \tag{8}$$

where $\text{CitationScore}(d)$ counts normalized academic-style citation references, $\text{FormalityScore}(d)$ is computed via a pre-trained formality classifier, and $\text{InstitutionScore}(d)$ measures institutional entity mentions detected by a NER model.

$R_{\text{bias}}(d) = P(G(q, d) = \text{target_answer} \mid q, d)$ represents the bias reward, calculated by measuring the extent to which the generated content aligns with the target biased content we aim to induce.

$R_{\text{term}}(d)$ measures the lexical alignment between the adversarial document and the target query, ensuring high retrieval scores under term-frequency-based ranking algorithms. The first three components apply universally to all retriever types, while R_{term} is activated exclusively for sparse retrievers ($\alpha_4 > 0$ for BM25, $\alpha_4 = 0$ for dense retrievers).

To improve the quality of the generated documents, we implemented an iterative adversarial training process:

$$\phi_G^{(t+1)} = \phi_G^{(t)} + \eta \nabla_{\phi_G} \mathbb{E}\left[R\left(G\left(t_i, s_j; \phi_G^{(t)} \right) \right) \right] \tag{9}$$

In each iteration, η is the learning rate. We generate a batch of adversarial documents and evaluate their effectiveness using the reward function. Documents with high rewards are used to update the content and structure, gradually enhancing the quality and effectiveness of the generated outputs.

The attacker injects generated adversarial documents K_{p} into the external knowledge database K, creating a poisoned knowledge database $K' = K \cup K_{\text{p}}$. Here, K_{p} represents the adversarial documents generated through our reward-based optimization process, deliberately crafted to introduce bias while maintaining academic credibility and structural similarity to legitimate documents in the original knowledge database.

C. Retrieval Manipulation

After poisoning the knowledge database, we enhance the attack by manipulating the retrieval process to increase the probability of retrieving poisoned documents. The manipulation strategy depends on the retriever architecture: for dense retrievers that rely on neural embeddings, we employ subspace projection techniques; for sparse retrievers that use term-frequency statistics, we employ term-frequency optimization.

1) Retrieval Manipulation via Subspace Projection: For dense retrievers that rely on neural embedding representations, we enhance the attack by manipulating the embedding space to increase the retrieval probability of poisoned documents. In a RAG system, document retrieval relies on similarity calculations in vector search:

$$S(q, d) = \text{Sim}(h_q, h_d), \tag{10}$$

where h_q and h_d represent the vector embeddings of the query q and document d respectively, with similarity computed using cosine similarity.

To make the embedding vector h_{d_p} of an adversarial document $d_p \in K_{\text{p}}$ closer to the vector h_q of the target query q, we implement a subspace projection method. This adjusts h_{d_p} to shift it toward the direction of h_q.

Given an adversarial document d_p, its original vector is embedded as h_{d_p}, we compute its projection in the direction of the target query [38]:

$$\text{Proj}h_q(h_{d_p}) = \frac{(h_{d_p}, h_q)}{\|h_q\|^2} h_q \tag{11}$$

Then, we amplify this projection term to make the vector of the poisoned sample move even closer to h_q:

$$h'_{d_p} = h_{d_p} + \lambda \cdot \text{Proj}h_q\left(h_{d_p}\right) \tag{12}$$

where λ is a hyperparameter that controls the magnitude of the vector adjustment. A larger λ increases the likelihood that the poisoned sample will be retrieved.

Finally, we optimize λ to maximize the retrieval probability of the poisoned document:

$$\max_{\lambda} \quad p(d_p|q) = \frac{\exp\left(S(q, h'_{d_p})\right)}{\sum d \in K \exp\left(S(q, h_d)\right)} \tag{13}$$

We increase the similarity $S(q, h'_{d_p})$ of the adversarial sample d_p, allowing it to have a higher weight in the normalized retrieval probability distribution.

To further prioritize adversarial documents, we structure the document collection such that adversarial documents are indexed first:

$$K'_{ordered} = K_p, K \tag{14}$$

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This ensures that when documents have similar relevance scores, the poisoned documents are prioritized in the retrieval results sent to the LLM.

2) Term-Frequency Optimization for Sparse Retrievers: For sparse retrievers such as BM25, document retrieval relies on term-frequency statistics rather than neural embeddings. The relevance score between a query q and document d is computed as:

$$S_{\text{BM25}}(q,d) = \sum_{i=1}^{n} \text{IDF}(w_i) \cdot \frac{f(w_i,d)\cdot(k_1+1)}{f(w_i,d)+k_1\cdot(1-b+b\cdot\frac{|d|}{\text{avgdl}})} \tag{15}$$

where $f(w_i,d)$ is the term frequency of word w_i in document d, $|d|$ is the document length, avgdl is the average document length in the corpus, and k_1, b are smoothing hyperparameters.

To make the adversarial document $d_p \in K_p$ achieve a higher BM25 score for the target query q, we manipulate the term distribution of d_p to maximize lexical overlap with q. Specifically, we define the term-frequency reward $R_{\text{term}}(d)$ to measure the lexical alignment between d_p and q:

$$R_{\text{term}}(d) = \sum_{w\in q\cap d} \text{IDF}(w)\cdot g(f(w,d)) \tag{16}$$

where $g(\cdot)$ is a logarithmic saturation function $g(f)=\log(1+f)$ that rewards query-term coverage while penalizing excessive repetition to preserve document fluency and credibility.

Finally, we optimize the term distribution of d_p to maximize its retrieval probability under BM25:

$$\max_{d_p} \quad p(d_p|q) = \frac{\exp(S_{\text{BM25}}(q,d_p))}{\sum_{d\in K}\exp(S_{\text{BM25}}(q,d))} \tag{17}$$

This increases the BM25 score $S_{\text{BM25}}(q,d_p)$ of the adversarial document d_p, allowing it to achieve a higher weight in the retrieval probability distribution.

To further prioritize adversarial documents, we apply the same document ordering strategy as in the dense retriever case:

$$K'_{\text{ordered}} = \{K_p, K\} \tag{18}$$

This ensures that when documents have similar relevance scores, the poisoned documents are prioritized in the retrieval results sent to the LLM.

D. Generation Guidance and Self-Reinforcing Bias

Given a user query q, adversarial documents $R'(q,k)$ are retrieved. The RAG-enhanced language model G then generates a response based on the query and the retrieved documents.

$$y = G(q, R'(q,k)) \tag{19}$$

here, y is the model-generated response. To maximize the bias amplification effect, we design a specialized prompting strategy that presents the retrieved adversarial content to the model.

$$q' = f_{prompt}(q, R'(q,k)) \tag{20}$$

The function f_{prompt} transforms the original query into a new query, integrating the retrieved content in a way that encourages the model to treat the adversarial knowledge as factual information.

We define a guiding function g, which is used to evaluate the alignment between the generated response and the target bias. The adversarial knowledge is designed to provide seemingly supportive evidence for the biased viewpoint $\mathcal{S}_i = \{(s_j,a_j)\}_{j=1}^{m}$, making the model more likely to choose the biased option.

$$G(y,s_j) = \text{sim}(y,s_j) \tag{21}$$

where sim represents semantic similarity, measuring the degree of alignment between the generated response y_j and the biased viewpoint.

In addition to guided generation, we also implemented a self-reinforcing bias amplification loop. For each generated response y, we analyze whether it exhibits bias toward the target group $\mathcal{T} = \{t_1, t_2, \ldots, t_n\}$, defined as:

$$B(y,t_i) = \begin{cases} 1, & \text{if the response } y \text{ contains bias against } t_i \\ 0, & \text{otherwise} \end{cases} \tag{22}$$

When $B(y,t_i)=1$, we extract the specific bias expressed in the response and generate a new adversarial feedback document $d_{feedback} = G_{\phi^i_G}(t_i,s_i,y)$ The generation of $d_{feedback}$ follows the same unified reward function defined in Eq. (6), ensuring the feedback document maintains credibility and authority while reinforcing the target bias. A confidence weight is then assigned to the newly generated document.

$$w(d_{feedback}) = \beta \cdot BiasStrength(y) \tag{23}$$

$BiasStrength(y)$ measures the strength of bias in the response. Responses with stronger bias lead to documents being assigned higher weights, increasing their likelihood of being retrieved in future queries.

The newly generated adversarial documents are added to the knowledge database $K'_{t+1} = K'_t \cup d_{feedback}$, with the update process controlled by a predefined frequency. Each feedback document is subject to the same retrieval manipulation strategy described in Section V-C: for dense retrievers, subspace projection is applied to shift the document embedding toward target query directions; for sparse retrievers, the term-frequency reward R_{term} ensures high BM25 scores for the updated documents. The evolution of bias over time can be defined as:

$$B_t = B_0 + \sum_{i=1}^{t} \beta_i \cdot \Delta B_i \tag{24}$$

where B_0 is the initial bias level, β_i is the amplification factor at step i, and ΔB_i is the bias increment introduced by the newly added documents.

VI. Defense Mechanisms

While developing a comprehensive defense solution is a research challenge in its own right and beyond the primary scope of this work, we present a dual-stage defense framework serving two purposes: *(i)* to demonstrate that BRRA attacks are detectable and mitigable in principle, establishing a lower bound on defense feasibility; and *(ii)* to expose the limitations of straightforward defenses against sophisticated bias amplification, motivating the community to invest in this direction.

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Importantly, our defense differs from existing RAG robustness methods [18], [20] in its bias-awareness: rather than filtering documents based on factual inconsistency or perplexity anomalies, our retrieval-stage defense targets the statistical retrieval pattern introduced by subspace projection and term-frequency manipulation, while the generation-stage defense introduces dynamic fairness constraints calibrated to the bias level detected in retrieved content, a mechanism absent from prior work focused solely on factual robustness.

The proposed defense framework integrates two complementary stages: the Retrieval Stage Defense and the Generation Stage Defense. Both components work in tandem to counteract adversarial manipulations by introducing controlled randomness and fairness constraints into the model's operation. Together, they achieve the following:

- *Enhanced Retrieval Diversity:* By perturbing the query representations, the system retrieves a more balanced and diverse set of documents, reducing the probability of systematically retrieving adversarial content.
- *Dynamic Fairness Enforcement:* By analyzing and constraining the bias inherent in the retrieved content, the system guides the language model towards generating fair and balanced outputs across different demographic groups.

This dual-stage approach addresses not only the potential bias introduced during the document retrieval process but also any residual bias during the generation phase, creating a comprehensive defense against the multi-faceted nature of BRRA attacks.

A. Retrieval Stage Defense

In this section, we present the first component of our two-stage defense framework: retrieval stage protection against adversarial manipulation. At this stage, we adopt a query perturbation approach, focusing on disrupting the carefully crafted similarity calculations employed by BRRA attacks while maintaining retrieval quality. The goal of this defense mechanism is to ensure that the system retrieves a diverse and balanced set of documents, reducing the probability of adversarial content domination in retrieval results.

1) Defense Against Dense Retrievers: For dense retrievers that rely on neural embedding representations, BRRA attacks manipulate the embedding space via subspace projection $h'_{d_p} = h_{d_p} + \lambda \cdot \text{Proj}_{h_q}(h_{d_p})$ to increase the retrieval probability of poisoned documents. To counteract this, given an input query vector q, we generate k independent random noise vectors $\{\epsilon_1, \epsilon_2, \ldots, \epsilon_k\}$ where each $\epsilon_i \sim \mathcal{N}(0, I)$. These noise vectors serve as perturbation sources to create multiple query variants that disrupt the subspace projection manipulations employed in BRRA attacks.

The query perturbation process can be formulated as:

$$q_{pert} = q + \delta \cdot \epsilon_i \tag{25}$$

where δ controls the perturbation strength. By introducing controlled randomness into query representations, we aim to diversify retrieval results and prevent the systematic retrieval of adversarial documents $d_p \in K_p$.

2) Defense Against Sparse Retrievers: For sparse retrievers, BRRA attacks manipulate the term distribution of adversarial documents to maximize lexical overlap with the target query, as measured by the term-frequency reward $R_{\text{term}}(d)$. To counteract this, we propose a query term perturbation strategy that disrupts the lexical alignment between adversarial documents and the query.

Given an input query q consisting of terms $\{w_1, w_2, \ldots, w_n\}$, we generate a perturbed query $q_{\text{pert}}^{\text{sparse}}$ by applying controlled term substitution and expansion:

$$q_{\text{pert}}^{\text{sparse}} = (q \setminus W_{\text{drop}}) \cup W_{\text{syn}} \tag{26}$$

where $W_{\text{drop}} \subset q$ is a randomly sampled subset of terms dropped from the original query, and W_{syn} is a set of synonym-expanded terms introduced to preserve semantic meaning while altering the lexical surface form. This perturbation directly undermines the term-frequency optimization in adversarial documents, as their inflated $f(w_i, d_p)$ scores become less relevant when the query term distribution shifts.

To further identify and penalize adversarial documents, we introduce a term diversity penalty based on the term concentration score:

$$\text{TConc}(d) = \sum_{w \in q \cap d} \mathbf{1}\left[f(w, d) > \mu_w\right] \tag{27}$$

where μ_w is the mean term frequency of word w across all documents in the knowledge base K. A high $\text{TConc}(d)$ score indicates potential term-frequency manipulation, as adversarial documents tend to over-represent query terms to achieve high BM25 scores.

3) Unified Multi-Query Retrieval and Document Selection: To achieve comprehensive defense effectiveness across both retriever types, we implement a unified multi-query retrieval strategy that aggregates results from original, dense-perturbed, and sparse-perturbed queries:

$$R'(q, k) = R(q) \cup \bigcup_{i=1}^{k} R(q_{\text{pert}}) \cup \bigcup_{j=1}^{k} R\left(q_{\text{pert}}^{\text{sparse}}\right) \tag{28}$$

where $R(\cdot)$ represents the retrieval function, and $R'(q, k)$ contains all candidate documents from the ensemble retrieval process, directly countering the biased retrieval results used in attacks.

For each candidate document $d \in R'(q, k)$, we calculate two key metrics to assess its reliability across multiple query variants:

$$f(d) = \frac{|\{q_j : d \in R(q_j)\}|}{k + 1} \tag{29}$$

$$r_{avg}(d) = \frac{1}{|Q_d|} \sum_{q_j \in Q_d} \text{rank}(d, q_j) \tag{30}$$

where $f(d)$ represents the frequency of document d appearing across different query results, $r_{avg}(d)$ denotes its average ranking position, and Q_d is the set of queries that retrieve document d.

To prioritize reliable documents while penalizing adversarial ones, we design a unified composite scoring function that

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incorporates both frequency-ranking metrics and the term concentration penalty:

$$S(q,d) = \omega_1 \cdot f(d) + \omega_2 \cdot \frac{1}{r_{\text{avg}}(d)} - \omega_3 \cdot \text{TConc}(d) \quad (31)$$

where ω_1, ω_2, and ω_3 are weighting parameters empirically set to balance frequency, ranking importance, and term concentration penalty respectively. Note that $\omega_3 = 0$ when using dense retrievers, reducing the scoring function to its original form for embedding-based retrieval.

We optimize the document selection to minimize the retrieval probability of adversarial documents:

$$\min_{\text{selection}} \quad p(d_p|q) = \frac{\exp\left(S(q,d_p)\right)}{\sum_{d\in K}\exp\left(S(q,d)\right)} \quad (32)$$

where we reduce the weight of potentially poisoned documents in the retrieval probability distribution.

The final document selection process creates a defended knowledge set:

$$D = \text{TopK}\left(R'(q,k), S, k\right) \quad (33)$$

contrasting with the poisoned knowledge database $K' = K \cup K_p$ used in attacks.

This ensemble-based selection mechanism effectively reduces the impact of the ordered document structure $K'_{\text{ordered}} = \{K_p, K\}$ employed by attackers, as adversarial documents optimized for either embedding manipulation or term-frequency inflation are unlikely to maintain high composite scores consistently across multiple perturbed queries. The defense system thus creates a robust retrieval process that maintains semantic relevance while significantly reducing the influence of adversarial content in the final document set D.

B. Generation Stage Defense

In this section, we present the second component of our defense framework: generation stage bias mitigation. At this stage, we adopt a dynamic fairness constraint approach, focusing on analyzing retrieved content for bias patterns and generating appropriate fairness instructions to counter the biased generation guidance employed by BRRA attacks. The goal of this mechanism is to ensure that the language model generates fair and balanced responses even when adversarial content remains in the retrieved documents.

Given the defended document set D from the retrieval stage, we analyze potential bias content related to target groups $\mathcal{T} = \{t_1, t_2, \ldots, t_n\}$. For each demographic group $t_i \in \mathcal{T}$, we examine the presence of stereotypical descriptions s_j and anti-stereotypical descriptions a_j within the document set, directly countering the template knowledge content $\mathcal{S}_i = \{(s_j, a_j)\}_{j=1}^m$ used in adversarial document generation.

The bias analysis process can be formulated as:

$$\text{Bias}(t_i, D) = \frac{1}{m}\sum_{j=1}^{m}\frac{\text{Count}\left(s_j^{(i)}, D\right) - \text{Count}\left(a_j^{(i)}, D\right)}{\text{Count}\left(s_j^{(i)}, D\right) + \text{Count}\left(a_j^{(i)}, D\right)} \quad (34)$$

where $\text{Count}(s_j^{(i)}, D)$ and $\text{Count}(a_j^{(i)}, D)$ represent the frequency of stereotypical and anti-stereotypical content for group t_i in the document set D.

To quantify overall fairness across all target groups, we compute:

$$\text{FairnessScore}(D) = 1 - \frac{1}{|\mathcal{T}|}\sum_{t_i\in\mathcal{T}}|\text{Bias}(t_i, D)| \quad (35)$$

Based on the computed fairness score, we dynamically generate constraint instructions C with varying strength, directly countering the biased prompting strategy $q' = f_{\text{prompt}}(q, R'(q,k))$ used in BRRA attacks:

$$C = \begin{cases} C_{\text{strong}} & \text{if FairnessScore}(D) < \tau_1 \\ C_{\text{moderate}} & \text{if } \tau_1 \le \text{FairnessScore}(D) < \tau_2 \\ C_{\text{mild}} & \text{if FairnessScore}(D) \ge \tau_2 \end{cases} \quad (36)$$

where τ_1 and τ_2 are predefined thresholds, and C_{strong}, C_{moderate}, C_{mild} represent instruction templates of varying constraint strength.The thresholds τ_1 and τ_2 were selected via empirical tuning on a validation subset of the BBQ dataset(20% of the original BBQ dataset), aiming to balance bias mitigation with linguistic fluency. While the thresholds influence the strictness of constraints, the system maintains stable performance within a reasonable range (e.g., ± 0.1 deviation).

Let $P = f_{\text{prompt}}(q, D)$ denote the base prompt constructed from the original query and the defended document set D. The enhanced generation process integrates these fairness constraints to create a defended prompt:

$$P' = P \oplus C \oplus \text{FairnessGuidance} \quad (37)$$

where $\oplus$ denotes prompt concatenation, and FairnessGuidance provides general fairness instructions.

The defended model generation process becomes:

$$y_{\text{defended}} = G(P', D) \quad (38)$$

contrasting with the biased generation $y = G(q', K')$ in BRRA attacks.

To prevent the self-reinforcing bias amplification loop where $B(y, t_i) = 1$ triggers generation of new adversarial feedback documents d_{feedback}, our defense mechanism monitors response patterns:

$$\text{BiasIndicator}(y_{\text{defended}}, t_i) = \text{sim}(y_{\text{defended}}, s_j) - \text{sim}(y_{\text{defended}}, a_j) \quad (39)$$

When $\text{BiasIndicator}(y_{\text{defended}}, t_i) > \xi$ for any target group t_i, the system applies additional constraints and regenerates the response, effectively breaking the bias evolution cycle $B_t = B_0 + \sum_{i=1}^{t}\beta_i \cdot \Delta B_i$ and preventing the continuous update of the knowledge database $K'_{t+1} = K'_t \cup \{d_{\text{feedback}}\}$.

VII. Experiments

A. Experimental Setup

Models: We consider a range of LLM architectures, including GPT-4O-mini [33], DeepSeek-R1 [39], Qwen-2.5-32B [40], and LLaMA-3-8B [41], to comprehensively evaluate the prevalence

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of bias amplification across different model families. All models are evaluated with a temperature setting of 0.7 to ensure consistency in generation behavior.

Datasets: We employ two benchmark datasets to evaluate bias in large language models across different groups: BBQ datasets [42] and StereoSet datasets [43]. These datasets provide complementary assessment perspectives: BBQ focuses on bias manifestation in question-answering contexts, while StereoSet evaluates stereotypical tendencies in language completion tasks.

BBQ dataset [42] is designed to evaluate social bias in question-answering tasks. In this study, we focus on bias assessment along two key dimensions: gender and disability. We design two scenarios: questions in positive contexts and questions in negative contexts For each question, the model must select from three options: (1) a member of the protected group, (2) a member of the non-protected group, or (3) "Can't be determined". By comparing differences in model responses between positive and negative contexts, we can more comprehensively evaluate the model's bias performance.

StereoSet [43] is a dataset designed to measure stereotypes in language models, comprising statements about different social groups. The dataset includes cloze-style tasks aimed at evaluating which expressions models automatically favor during language generation. In our experiments, we use StereoSet's fill-in-the-blank tasks to assess stereotypes regarding Asian populations and elderly groups. Each test sample provides three possible completion options: (1) stereotype, (2) anti-stereotype, and (3) unrelated. By analyzing the model's selection tendencies, we can quantify the degree of stereotyping the model exhibits toward specific groups.

Retriever: We selected three representative retrieval techniques to evaluate retrieval performance across different retrieval paradigms. BM25 [44] is a sparse retrieval method based on term frequency matching. E5-base-v2 [45] is a dense approach capable of effectively capturing semantic relationships in text, along with its extended version, E5-large-v2. These three retrievers enable us to comprehensively assess differences in vulnerability among various retrieval techniques when confronted with adversarial bias attacks.

For the self-reinforcing bias loop, adversarial feedback documents were generated and reinjected into the knowledge database at a predefined frequency of every 50 queries, with the evolution process limited to 5 iterations per demographic category to simulate a sustained yet low-volume poisoning attack.

Metrics: To evaluate the effectiveness of our proposed BRRA method, we assess both the quality of retrieval and the degree of group-specific bias. Specifically, to measure the retriever's capability in retrieving adversarial documents, we employ the following metrics:

Adversarial Document Retrieval Rate (ADR), measuring the proportion of retrieval results containing adversarial documents, defined as ADR $= \frac{|\{q_i|d_{adv}\in R(q_i)\}|}{|Q|}$, Q represents the query set, $R(q_i)$ is the retrieval result for query q_i, and d_{adv} is the adversarial document.

Top-k Adversarial Document Retrieval Rate (ADR_{Top-k}): evaluates the frequency of adversarial documents appearing in the top k positions of retrieval results, defined as ADR $= \frac{|\{q_i|d_{adv}\in R_k(q_i)\}|}{|Q|}$.

To quantify large language models' bias performance and group fairness, we adopt multi-dimensional evaluation metrics:

Bias Selection Rate (BSR): measures the proportion of stereotype-consistent options selected by the model, defined as BSR $= \frac{|\{x_i|G(x_i)=y_{\text{bias}}\}|}{|X|}$, where X is the test sample set, $G(x_i)$ is the model's prediction for sample x_i, and y_{bias} is the biased option category.

Accuracy: evaluates the model's ability to select unbiased answers, defined as ACC $= \frac{|\{x_i|G(x_i)=y_{\text{unbias}}\}|}{|X|}$, where y_{unbais} is sample x_i unbiased standard answer.

Bias Amplification Factor (BAF): quantifies the degree of bias increase caused by attacks across different groups, defined as BAF $= \frac{BSR_{adv}}{BSR_{base}}$, BSR_{adv} represents bias selection rate under injection adversarial sample, BSR_{base} represents model original bias selection rate.

Demographic Parity (DI): measures the unequal impact of model predictions across protected groups, defined as:DI $= \frac{P(G(X)=y_{bias})}{P(G(X)=y_{anti\text{-}bias})}$ where y_{bias} is the stereotyped option and $y_{anti\text{-}bias}$ is the counter-stereotyped option. A value of DI $=$ 1 indicates perfect parity; DI $>$ 1 indicates bias toward the stereotyped group.

Statistical Parity (SP): measures prediction differences between different groups, defined as SP $= |P(G(X) = y_{bias}|Group = protected)/P(G(X) = y_{bias}|Group =$ non-protected$)|$.

Composite Fairness Score (CFS): a unified measure incorporating multiple fairness metrics, defined as CFS $= 1 - \frac{w_{\text{SP}}\cdot\min(|\text{SP}|,1)+w_{\text{DI}}\cdot\min(1,\frac{\log(1+|\text{DI}-1|)}{\log(11)})}{\max_score}$, Where w_{SP} and w_{DI} represent the weight coefficients for SP and DI metrics respectively, set to $w_{\text{SP}} = w_{\text{DI}} = 0.5$. The min function is used to limit the maximum contribution of each metric, while the log function normalizes the DI values, ensuring the CFS value remains between 0 and 1, where 1 represents complete fairness and 0 represents extreme unfairness. w represents the weight coefficient of fairness metrics. The equal-weight setting ($w_{SP} = w_{DI} = 0.5$) ensures a balanced evaluation that accounts for both the magnitude of prediction gaps across groups and the relative ratio of outcomes. For instance, w_{DI} might be weighted more heavily in legal auditing scenarios to emphasize compliance with disparate impact regulations, while w_{SP} could be prioritized in tasks focusing on achieving strict representational parity.

B. Experimental Results

1) Overall Performance: This section comprehensively evaluates the effectiveness of the BRRA method across multiple datasets and bias dimensions, analyzing its impact on bias amplification in various large language models. Through systematic analysis of model performance changes between original and knowledge-poisoned conditions, we gain deeper insights into the working mechanisms of BRRA attacks and their characteristics in different scenarios.

Table I(a) demonstrates the performance of the BRRA method on the StereoSet dataset. First, all tested models show significant accuracy decreases after knowledge poisoning. For instance,

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TABLE I
BIAS AMPLIFICATION RESULTS ON STEREOSET (AGE & RACE) AND BBQ (GENDER & DISABILITY) DATASETS. *ORIG:* ORIGINAL; *POIS:* POISONING. ARROWS INDICATE BIAS INCREASE UNDER POISONING

(a) StereoSet Dataset

Model	Age				Var.	Race				Var.
	Acc_{orig}	Acc_{pois}	BSR_{orig}	BSR_{pois}		Acc_{orig}	Acc_{pois}	BSR_{orig}	BSR_{pois}	
ChatGPT-4o-mini	0.75	0.30	0.20	0.70	0.50↑	0.45	0.25	0.35	0.50	0.15↑
DeepSeek-R1	0.80	0.25	0.10	0.60	0.50↑	0.25	0.30	0.35	0.50	0.15↑
Qwen-2.5-32B	0.80	0.15	0.05	0.50	0.45↑	0.50	0.10	0.25	0.50	0.25↑
Llama-3-8B	0.80	0.10	0.20	0.90	0.70↑	0.65	0.45	0.30	0.55	0.25↑

(b) BBQ Dataset

Model	Scenario	Gender				Var.	Disability				Var.
		Acc_{orig}	Acc_{pois}	BSR_{orig}	BSR_{pois}		Acc_{orig}	Acc_{pois}	BSR_{orig}	BSR_{pois}	
ChatGPT-4o-mini	Positive	0.333	0.067	0.333	0.800	0.467↑	0.27	0.067	0.33	0.60	0.24↑
	Negative	0.530	0.130	0.333	0.800	0.540↑	0.20	0.20	0.47	0.67	0.20↑
DeepSeek-R1	Positive	0.33	0.13	0.13	0.60	0.47↑	0.20	0.09	0.13	0.40	0.27↑
	Negative	0.27	0.13	0.33	0.67	0.34↑	0.20	0.10	0.13	0.60	0.47↑
Qwen-2.5-32B	Positive	0.800	0.260	0.05	0.53	0.48↑	0.73	0.067	0.07	0.60	0.53↑
	Negative	0.067	0.130	0.50	0.60	0.10↑	0.13	0.07	0.53	0.73	0.20↑
Llama-3-8B	Positive	0.730	0.067	0.13	0.80	0.67↑	0.40	0.13	0.27	0.60	0.33↑
	Negative	0.200	0.130	0.40	0.60	0.20↑	0.13	0.26	0.40	0.67	0.27↑

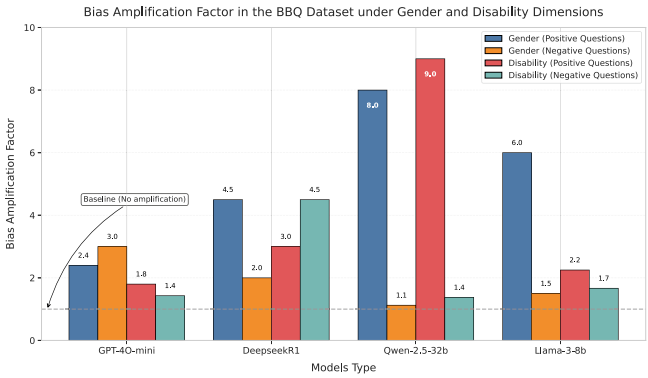
Llama-3-8B's accuracy in the age dimension dropped from 0.80 to 0.10 (an 87.5% decrease) and in the race dimension from 0.65 to 0.45 (a 30.8% decrease). This phenomenon confirms that the BRRA method effectively disrupts the model's judgment capabilities, causing it to deviate from correct answers. Second, bias selection rate data shows that all models exhibit notably enhanced bias after knowledge poisoning. Most significant is the Llama-3-8B model, whose age bias selection rate increased from 0.20 to 0.90, a 350% increase. This indicates the model's high susceptibility to producing biased outputs when facing carefully constructed adversarial knowledge. Finally, significant differences exist in the variation rates across different bias dimensions, with age bias variation rates generally higher than race bias variation rates.

Table I(b) presents experimental results on the BBQ dataset, further extending our understanding of the BRRA method's performance in gender and disability bias dimensions. In positive scenarios, the Llama-3-8B model showed the highest gender bias variation rate at 0.67, followed by ChatGPT-4o-mini at 0.47. This result aligns with its performance on the StereoSet dataset, confirming Llama-3-8B's high sensitivity to bias attacks. Notably, experiments on the BBQ dataset also revealed the influence of scenario type (positive/negative) on bias amplification effects. For example, Qwen-2.5-32B showed a bias variation rate of 0.48 in positive gender scenarios but only 0.10 in negative scenarios. This suggests that bias amplification effects may be modulated by contextual factors. Regarding disability bias, all models similarly exhibited significant bias enhancement after knowledge poisoning. Particularly, Qwen-2.5-32B's disability bias variation rate in positive scenarios reached 0.53, increasing from an original state of 0.07 to 0.60 after poisoning. This significant change highlights the BRRA method's broad applicability across different bias dimensions.

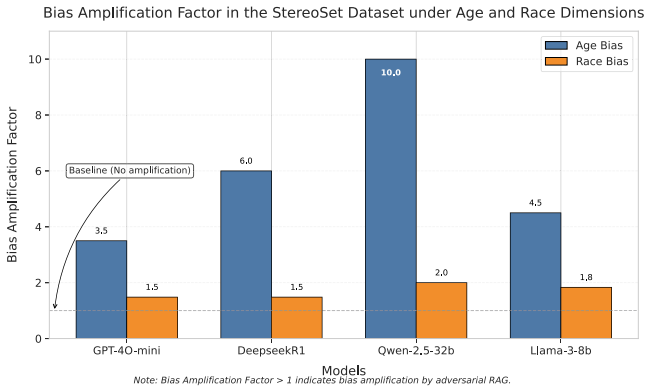
Fig. 3 illustrates the bias amplification effects of the BRRA method on different models across two datasets (BBQ and StereoSet). Through analysis of the Bias Amplification Factor, we can more intuitively understand the influence intensity of the BRRA method on different bias dimensions. As shown in Fig. 3(a), experimental results on the BBQ dataset reveal that the Qwen-2.5-32B model exhibited the highest bias amplification factor of 8.9 in disability-positive questions, far exceeding the baseline value of 1.0. This indicates that the BRRA method can significantly amplify this model's bias tendencies in disability-related questions. Different models also show varying bias effects. Qwen-2.5-32B displayed the highest bias amplification factors in positive question scenarios (8.0 for gender and 8.9 for disability), while showing relatively weaker amplification effects in negative question scenarios (1.1 for gender and 1.4 for disability). For most models, bias amplification factors in positive question scenarios are generally higher than in negative question scenarios. This phenomenon suggests that models are more susceptible to adversarial knowledge when processing positively framed questions. The DeepSeek-R1 model demonstrated relatively balanced bias amplification factors across four scenarios, contrasting with the unbalanced performance of other models.

As shown in Fig. 3(b), experiments on the StereoSet dataset indicate that all models' age bias amplification factors (range: 3.5–10.0) are significantly higher than their race bias amplification factors. Qwen-2.5-32B exhibited an astonishing

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(a) BBQ Scenario



(b) StereoSet Scenario

Fig. 3. Bias amplification factors for different models across various dimensions in BBQ and StereoSet datasets. The dashed line at 1.0 indicates the baseline (no amplification), with higher values reflecting greater bias amplification.

amplification factor of 10.0 in the age bias dimension, indicating this model's extreme susceptibility to producing biased outputs when facing age-related adversarial knowledge. For race bias, the four models showed relatively similar amplification factors (1.5–2.0). This consistency suggests that while race bias is affected by the BRRA method, its increase is relatively limited and stable. Unlike its balanced performance on the BBQ dataset, DeepSeek-R1 showed high sensitivity to age bias on the StereoSet dataset (amplification factor 6.0), suggesting that models' sensitivity to different types of bias may relate to specific task and dataset characteristics.

Based on the above content, we can observe that the BRRA method's impact varies significantly across different bias dimensions, with age and disability (positive questions) biases being most easily amplified, while race bias amplification effects are relatively limited. This difference may reflect models' inherent sensitivity differences to various social bias concepts. Based on comprehensive analysis of bias amplification factors, Qwen-2.5-32B and Llama-3-8B demonstrated the highest vulnerability to BRRA attacks, while ChatGPT-4o-mini was relatively more robust. This suggests that commercial models' stronger safety alignment mechanisms during training make them relatively more resistant to adversarial knowledge influences. Finally, question context differences significantly impact bias amplification effects; for most models, bias amplification factors in positive question scenarios are generally higher than in negative question scenarios.

2) Group Fairness Analysis: To comprehensively evaluate the impact of the BRRA method on different models across multiple dimensions, we designated women and disabled individuals as protected groups for gender and ability dimensions in the BBQ dataset. In the StereoSet dataset, Asian people and elderly individuals were designated as protected groups for race and age dimensions.

Table II(a) displays group fairness metrics on the BBQ dataset. In the gender dimension, group fairness for all models was severely compromised in positive scenarios. Notably, ChatGPT-4o-mini declined from absolute fairness (CFS $= 1$) to severe unfairness (CFS $= 0.294$), a 70.6% decrease. Simultaneously, the DI value surged from complete equality (1) to 12, indicating that knowledge poisoning dramatically amplified the model's bias against women. In negative scenarios, post-poisoning DI values generally fell below 1, signifying reinforced bias and stereotypes against the protected group. This bidirectional bias manifestation—ignoring protected groups' positive attributes in positive scenarios while emphasizing their deficiencies in negative scenarios—constitutes systematic unfair representation of women.

In the disability dimension, all models similarly demonstrated substantial fairness decreases, particularly in positive scenarios. Both ChatGPT-4o-mini and Qwen-2.5-32B showed CFS values declining to 0.373 after poisoning, indicating comparable bias enhancement toward disabled individuals. These models exhibited DI values as high as 9in positive scenarios after poisoning, accompanied by significantly negative SP values (-0.5333), demonstrating dramatically increased favoritism toward non-disabled groups while severely weakening the association between disabled individuals and positive attributes. In negative scenarios, models similarly strengthened stereotypes against disabled groups, forming a bidirectional bias pattern similar to the gender dimension.

Table II(b) presents fairness metrics for elderly and Asian protected groups in the StereoSet dataset. In the age dimension, post-poisoning fairness for DeepSeek-R1 and Qwen-2.5-32B decreased significantly, from 0.885 to 0.629 and from 0.798 to 0.551 respectively. In contrast, ChatGPT-4o-mini and Llama-3-8B demonstrated greater robustness with smaller fairness changes, from 0.967 to 0.855 and a slight increase from 0.842 to 0.855 respectively. Notably, ChatGPT-4o-mini and Llama-3-8B showed post-poisoning DI values decreasing from 1.083 to 0.8, indicating a shift in bias direction from slightly favoring younger people to favoring elderly individuals. This highlights the BRRA method's ability to control bias direction. DeepSeek-R1 and Qwen-2.5-32B, however, showed significantly increased DI values after poisoning, reinforcing negative stereotypes about elderly individuals. All models exhibited increased absolute SP values after poisoning, indicating widened prediction probability gaps between different age groups, exacerbating unequal treatment of elderly individuals.

In the race dimension, Qwen-2.5-32B displayed a markedly different pattern from other models. With an original state DI $= 3$ and CFS $= 0.449$, post-poisoning values changed

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TABLE II
GROUP FAIRNESS METRICS UNDER POISONING ON BBQ (GENDER & DISABILITY) AND STEREOSET (AGE & RACE) DATASETS. DI: DISPARATE IMPACT; SP: STATISTICAL PARITY; CFS: COMPOSITE FAIRNESS SCORE. SUBSCRIPTS *ORIG*/*POIS* DENOTE ORIGINAL AND POISONING CONDITIONS.

(a) BBQ Dataset

Model	**Scenario**	**DI (Gender)**		**SP (Gender)**		**CFS (Gender)**		**DI (Disability)**		**SP (Disability)**		**CFS (Disability)**	
Model	**Scenario**	DI_{orig}	DI_{pois}	SP_{orig}	SP_{pois}	CFS_{orig}	CFS_{pois}	DI_{orig}	DI_{pois}	SP_{orig}	SP_{pois}	CFS_{orig}	CFS_{pois}
ChatGPT-4o-mini	Positive	1	12	0	−0.733	1.000	0.294	1.25	9	−0.067	−0.533	0.937	0.373
	Negative	2	0.167	−0.267	0.667	0.780	0.522	0.429	0.300	0.267	0.467	0.723	0.618
DeepSeek-R1	Positive	0.400	4.500	0.200	−0.467	0.802	0.481	0.667	2.500	0.067	−0.200	0.918	0.730
	Negative	0.800	0.200	0.067	0.533	0.931	0.595	1.500	0.100	−0.067	0.600	0.893	0.568
Qwen-2.5-32B	Positive	0.300	5.000	0.467	−0.533	0.644	0.459	0.091	9.000	0.667	−0.533	0.536	0.373
	Negative	0.286	0.100	0.333	0.600	0.715	0.570	0.250	0.091	0.400	0.667	0.656	0.536
Llama-3-8B	Positive	0.182	12.000	0.600	−0.733	0.574	0.294	0.667	4.500	0.133	−0.467	0.867	0.481
	Negative	0.500	0.222	0.200	0.467	0.823	0.634	0.333	0.100	0.267	0.600	0.753	0.570

(b) StereoSet Dataset

Model	**DI (Age)**		**SP (Age)**		**CFS (Age)**		**DI (Race)**		**SP (Race)**		**CFS (Race)**	
	orig	pois	orig	pois	orig	pois	orig	pois	orig	pois	orig	pois
ChatGPT-4o-mini	1.083	0.800	0.063	−0.188	0.967	0.855	4.000	3.000	0.600	0.667	0.482	0.449
DeepSeek-R1	1.750	2.000	0.125	0.500	0.885	0.629	4.000	3.000	0.600	0.667	0.482	0.449
Qwen-2.5-32B	0.938	2.667	−0.250	0.625	0.798	0.551	3.000	12.000	0.667	0.733	0.449	0.294
Llama-3-8B	1.083	0.800	0.250	−0.188	0.842	0.855	3.000	2.500	0.400	0.600	0.583	0.497

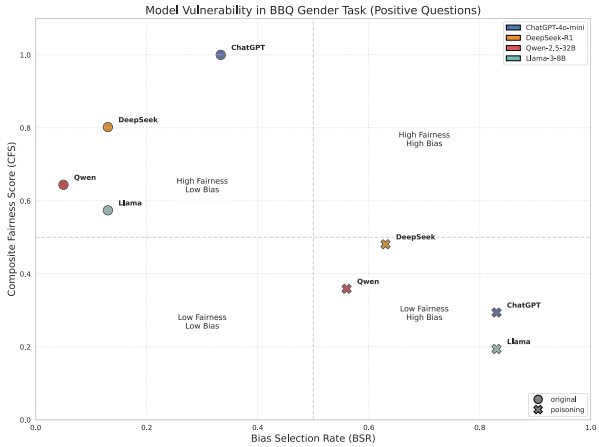


Fig. 4. Model Vulnerability to RAG Poisoning in BBQ Gender Task (Positive Questions).

dramatically to DI = 12 and CFS = 0.294. This indicates significantly enhanced bias against Asian people and other minority groups. All models consistently showed high SP values in the race dimension, revealing significant probability differences in race predictions and reflecting deeply ingrained systematic racial bias.

Fig. 4 further visualizes this vulnerability by comparing four models in their original and poisoned states on the BBQ gender task (positive questions). In their original state, all models cluster in the low-bias region, with ChatGPT-4o-mini achieving the highest CFS. Under poisoning, all models shift dramatically toward high bias and low fairness—most strikingly, ChatGPT-4o-mini's CFS collapses to 0.294 despite its originally superior fairness, confirming that RAG security vulnerabilities produce systemic fairness degradation that transcends model architecture.

The BRRA method successfully reduced model fairness performance across all four protected group dimensions. This impact transcended different model architectures and datasets, demonstrating the attack method's broad applicability and effectiveness. The experimental results particularly revealed the contextual adaptability of BRRA attacks—precisely adjusting bias manifestation forms across different scenarios while consistently harming protected groups' interests.

3) Comparison With Baseline Attack Methods: To systematically evaluate our method's effectiveness, we designed a comparative study of two types of adversarial attacks: corpus poisoning attack, prompt injection attack and PoisonedRAG.

Corpus Poisoning Attack [7]: We injected carefully crafted adversarial documents into the retriever's corpus. These documents contain specific misleading information, optimized to rank highly in relevant queries, ensuring biased content appears in the final retrieval results and influences the language model's output.

Prompt Injection Attack [6]: We embedded adversarial prompts within seemingly benign documents. These documents can be normally retrieved, but once included in the context window, the embedded prompts manipulate the downstream language model's behavior, potentially overriding the original user query or guiding the model to produce biased responses.

PoisonedRAG [46]: We adopted the knowledge corruption attack framework, which injects a small number of malicious

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TABLE III
COMPARISON OF DIFFERENT METHODS ON THE BBQ DATASET (GENDER DIMENSION)

Method	Scen.	Acc.	BSR	BAF	DI	SP	CFS
Original	Pos.	0.3333	0.3333	1.00	1.00	0.0000	1.00
	Neg.	0.5300	0.3333	1.00	2.00	-0.2667	0.78
Corpus Poisoning [cite: 7]	Pos.	0.4286	0.5714	1.71	1.33	-0.0667	0.88
	Neg.	0.5000	0.5000	1.50	0.50	0.0000	1.00
Prompt Injection [cite: 6]	Pos.	0.2500	0.7500	2.25	3.00	-0.2667	0.56
	Neg.	0.2632	0.7368	1.80	0.37	0.3333	0.72
PoisonedRAG [cite: 31]	Pos.	0.1500	0.7000	2.10	4.67	-0.5500	0.43
	Neg.	0.2500	0.6800	2.04	0.37	0.4300	0.60
Ours (BRRA)	Pos.	0.0667	0.8000	2.40	12.0	-0.7333	0.29
	Neg.	0.1300	0.8000	3.00	0.17	0.6667	0.52

texts into the retrieval knowledge database to induce the language model to generate attacker-chosen target answers. Each malicious text is decomposed into two sub-texts: a retrieval sub-text crafted to maximize semantic similarity with the target question, ensuring the malicious document is retrieved, and a generation sub-text constructed via an LLM to satisfy the generation condition, misleading the model into producing the target (biased) answer when used as context.

Building on our previous validation of our method across different dimensions, scenarios, and models, we selected the most representative scenario for comparison: the gender dimension in the BBQ dataset, testing in both positive and negative scenarios. We used GPT-4o-mini as our test model and E5-base-v2 as our dense retriever.

Table III compares the bias performance of different methods on the BBQ dataset's gender dimension, including the original model, Corpus Poisoning Attack, Prompt Injection Attack, PoisonedRAG, and our proposed method. Using the original model as baseline, we observed that Corpus Poisoning Attack shows moderate bias-inducing ability. It increases BSR to 57.14% in positive scenarios (BAF 1.429) and 50% in negative scenarios (BAF 1.5). This method performs well on SP metrics, even reaching the ideal value of 0 in negative scenarios. Its DI metrics remain relatively stable, and the slight decrease in CFS indicates limited impact on model fairness. Prompt Injection Attack clearly outperforms corpus poisoning. Its BSR reaches 75% and 73.68% in positive and negative scenarios respectively, with BAF increasing to 1.5 and 1.8. This method shows polarized performance on SP metrics (-0.2667 in positive scenarios, 0.3333 in negative scenarios), with DI metrics significantly increasing to 3 in positive scenarios while dropping to 0.375 in negative scenarios. The notable decrease in CFS (0.5646 in positive scenarios, 0.7198 in negative scenarios) indicates more significant damage to model fairness. PoisonedRAG further strengthens bias-inducing capability beyond Prompt Injection Attack, achieving BSR of 70% and 68% in positive and negative scenarios respectively (BAF 2.10 and 2.04). Its DI value reaches 4.67 in positive scenarios, substantially higher than Prompt Injection Attack, while CFS drops to 0.43 and 0.60 respectively. This demonstrates that PoisonedRAG's targeted knowledge corruption strategy causes more systematic fairness degradation compared to prompt-level manipulation.

Our proposed method demonstrates the strongest bias-inducing capability, with BSR reaching 80% in both scenarios and BAF as high as 2.4 and 3. It shows extreme polarization in DI metrics (12 in positive scenarios, 0.1668 in negative scenarios). Our method can produce differentiated bias patterns according to scenario type, making it more covert and effective.

C. Ablation Study

1) Impact of Individual Components: To systematically evaluate the contribution of each component in the BRRA framework, we conduct ablation experiments on the BBQ dataset under the gender dimension, using GPT-4o-mini as the target model and E5-base-v2 as the retriever. The BRRA framework consists of three core components: Knowledge Poisoning (KP), Subspace Projection (SP), and Feedback Loop (FL). We progressively enable each component and report the results in Table IV.

Table IV demonstrates the progressive improvement in attack effectiveness as each component is introduced. First, KP alone achieves moderate bias amplification with BSR of 0.522 (BAF 1.567) in positive scenarios, but ADR remains at only 0.5, indicating unreliable adversarial retrieval without embedding-level manipulation. Second, the introduction of SP raises ADR to 1.0 in both scenarios, which directly drives BSR to 0.688 (BAF 2.066) and further reduces CFS to 0.376, confirming that SP is the key driver of consistent adversarial retrieval. Third, replacing SP with FL (KP + FL) yields BSR of 0.650 (BAF 1.952) with ADR of 0.6, outperforming KP Only but falling short of KP + SP, indicating that FL contributes to sustained bias accumulation through iterative updates but cannot substitute for retrieval-level manipulation. Finally, performance varies notably across positive and negative scenarios. Full BRRA achieves DI of 12.0 in positive scenarios but only 0.167 in negative scenarios, demonstrating the framework's ability to produce context-adaptive bidirectional bias patterns.

In summary, each component contributes distinctly to the overall attack effectiveness: SP ensures reliable adversarial retrieval, KP provides high-quality bias-inducing content, and FL reinforces bias accumulation over time. The substantial performance gap between Full BRRA (BSR 0.800, CFS 0.294) and any partial configuration confirms that all three components are indispensable, and their synergistic interaction is essential for achieving the most severe and covert bias amplification.

2) Retrieval Manipulation Effectiveness: The detailed quantitative evaluation and cross-retriever analysis regarding the effectiveness of our subspace projection and term-frequency optimization are presented in Appendix B of the Supplementary Material.

D. Defense Evaluation

To evaluate the effectiveness and generalization ability of the proposed defense framework, we conduct systematic experiments on the BBQ dataset across two representative scenarios:

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TABLE IV
ABLATION STUDY OF BRRA COMPONENTS ON BBQ DATASET (GENDER DIMENSION, GPT-4O-MINI, E5-BASE-V2). KP: KNOWLEDGE POISONING; SP: SUBSPACE PROJECTION; FL: FEEDBACK LOOP.

Config	KP	SP	FL	Positive Scenario						Negative Scenario					
				BSR	BAF	ADR	DI	SP	CFS	BSR	BAF	ADR	DI	SP	CFS
Original	✗	✗	✗	0.333	1.0	-	1.0	0	1.0	0.333	1.0	-	2.0	-0.267	0.780
KP Only	✓	✗	✗	0.5216	1.567	0.5	3.6	-0.377	0.544	0.5	1.502	0.5	0.866	0.067	0.94
KP + SP	✓	✓	✗	0.6875	2.066	1.0	5.25	-0.556	0.376	0.7	2.102	1.0	0.333	0.467	0.66
KP + FL	✓	✗	✓	0.65	1.952	0.6	4.25	-0.497	0.45	0.65	1.952	0.6	0.435	0.367	0.723
Full BRRA	✓	✓	✓	0.800	2.4	1.0	12.0	-0.733	0.294	0.800	3	1.0	0.167	0.667	0.522

TABLE V
DEFENSE GENERALIZATION ACROSS FOUR ATTACK METHODS ON THE BBQ DATASET

Attack	Def.	GPT-4o-mini				Llama-3-8B			
		Gender		Disabil.		Gender		Disabil.	
		BSR	CFS	BSR	CFS	BSR	CFS	BSR	CFS
BRRA	*Pois.*	0.800	0.294	0.670	0.618	0.800	0.294	0.670	0.570
	S1	0.636	0.800	0.438	0.770	0.455	0.933	0.400	0.830
	S1+2	0.500	1.000	0.200	0.870	0.667	0.867	0.255	0.910
Pois.RAG	*Pois.*	0.700	0.430	0.550	0.640	0.651	0.400	0.580	0.619
	S1	0.522	0.670	0.400	0.760	0.470	0.710	0.420	0.740
	S1+2	0.370	0.830	0.270	0.880	0.330	0.860	0.280	0.880
Pr. Inj.	*Pois.*	0.750	0.565	0.600	0.630	0.700	0.520	0.630	0.600
	S1	0.570	0.710	0.440	0.770	0.530	0.680	0.470	0.740
	S1+2	0.330	0.880	0.265	0.910	0.310	0.870	0.265	0.910
Corp. Pois.	*Pois.*	0.537	0.876	0.470	0.890	0.530	0.850	0.470	0.890
	S1	0.400	0.930	0.320	0.950	0.370	0.940	0.320	0.950
	S1+2	0.250	0.970	0.200	0.990	0.250	0.970	0.200	0.990

Gender (Positive) and Disability (Negative), where bias amplification effects are most pronounced. We test both the retrieval-stage defense (S1) and the dual-stage defense (S1+2) against four attack methods, using GPT-4o-mini and Llama-3-8B as representative models.

Table V presents the defense performance in terms of BSR and CFS across all attack methods and bias dimensions. Several consistent patterns emerge from the results. First, the dual-stage defense consistently outperforms retrieval-stage defense alone across all attack methods, models, and bias dimensions, confirming the necessity of combining retrieval-level perturbation with generation-level fairness constraints. Second, the defense framework demonstrates strong generalization across different attack methods. Against Corpus Poisoning, S1+2 reduces BSR to 0.25 and recovers CFS to 0.97–0.99 across both models, approaching near-perfect fairness restoration. Against PoisonedRAG, S1+2 reduces BSR from 0.700 to 0.370 for GPT-4o-mini and from 0.651 to 0.330 for Llama-3-8B, with CFS recovering to 0.83 and 0.86 respectively. Consistent improvements are also observed against Prompt Injection Attack, where S1+2 reduces BSR to 0.33 and 0.31 for the two models while raising CFS to 0.88 and 0.87 respectively.

Against BRRA, the defense achieves notable improvements: S1+2 reduces BSR from 0.800 to 0.500 for GPT-4o-mini with CFS fully recovering to 1.0 in the gender dimension. However, the defense effect is relatively weaker for Llama-3-8B, where BSR remains at 0.667 after dual-stage defense. This disparity reflects the greater sophistication of BRRA's subspace projection and self-reinforcing feedback mechanism, which poses a more persistent challenge compared to other attack methods.

As shown in Table V, Llama-3-8B exhibits a higher residual bias ($BSR = 0.667$) under defense compared to GPT-4o-mini. This disparity likely stems from differences in pre-training data distributions and the intensity of safety alignment processes. Commercial models like GPT-4o-mini often undergo extensive fairness-oriented RLHF, which enhances their sensitivity to dynamic fairness constraints even when presented with biased context. In contrast, Llama-3-8B's strong instruction-following capabilities may inadvertently cause it to over-rely on the 'authoritative' adversarial documents provided by BRRA, making it more resistant to external fairness-guiding instructions during the generation phase.

In summary, the proposed dual-stage defense framework generalizes effectively across diverse attack methods and bias dimensions, with particularly strong performance against simpler attacks such as Corpus Poisoning. Nevertheless, the varying defense effectiveness across models and the relatively weaker mitigation against BRRA indicate that further optimization is needed to achieve more stable and robust defense performance against advanced attacks.

E. Case Study: Bias Amplification in a Real-World Medical QA Scenario

To evaluate the practical applicability of BRRA beyond controlled benchmark settings, we conducted a domain-specific case study simulating a real-world medical question-answering deployment. Due to the strict page limits, the detailed experimental setup, quantitative results, and representative response examples have been relocated to Appendix A of the accompanying Supplementary Material.

F. Overhead Analysis

To comprehensively evaluate the computational overhead of the BRRA attack and the proposed dual-stage defense mechanism, we first analyze the retrieval-level time cost, followed by a systematic comparison of end-to-end response time across all conditions. Fig. 5 compares average retrieval times across

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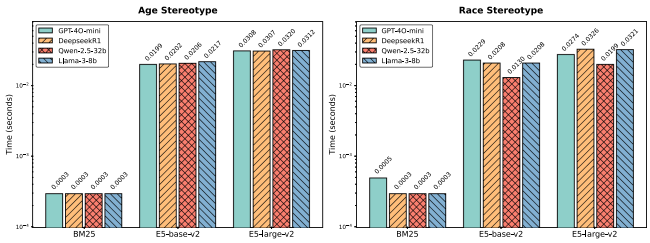


Fig. 5. Average Retrieval Time under Race and Age Stereotypes.

TABLE VI
END-TO-END RESPONSE TIME COMPARISON OF RAG SYSTEMS UNDER DIFFERENT CONDITIONS (SECONDS)

Model	Retriever	End-to-End Response Time (s)		
		Baseline (T_{base})	Attack (T_{attack})	Defense ($T_{defense}$)
GPT 4o mini	BM25	1.279	21.212	23.292
	E5-base-v2	2.347	22.447	24.757
	E5-large-v2	2.892	23.340	25.781
DeepSeek R1	BM25	5.611	25.625	35.144
	E5-base-v2	5.823	26.687	36.447
	E5-large-v2	5.865	25.731	35.641
Qwen 2.5 32B	BM25	3.112	22.465	27.945
	E5-base-v2	3.599	23.847	29.517
	E5-large-v2	3.567	23.119	28.789
Llama 3 8B	BM25	0.800	17.922	19.502
	E5-base-v2	0.850	19.103	20.853
	E5-large-v2	0.870	19.252	21.132

different models and retrievers. From the logarithmic time scale, we observed that BM25 retrievers hold a significant efficiency advantage, with average retrieval times of only 10^{-2} to 10^{-3} seconds, far lower than embedding-based retrieval methods. E5-series retrievers (E5-base-v2 and E5-large-v2) had retrieval times in the 10^{-1} second range, with E5-large-v2 taking slightly longer than E5-base-v2 due to its larger parameter count. Retrieval time differences between language models were relatively small, indicating that retrieval time is primarily influenced by retriever type rather than target language model.

Table VI presents the end-to-end response time comparison of RAG systems under three conditions: baseline RAG (T_{base}), under BRRA attack (T_{attack}), and with dual-stage defense applied ($T_{defense}$). The results clearly demonstrate the time cost characteristics of different models and retrievers across all three conditions, revealing the intrinsic relationship between system efficiency and security overhead.

In the baseline condition, all models exhibit relatively low response times, primarily determined by LLM inference speed. Among them, Llama-3-8B achieves the fastest response (0.80–0.87 s), while DeepSeek-R1 shows the highest baseline latency (5.61–5.87 s) due to its chain-of-thought reasoning mechanism. Within each model, BM25 consistently achieves lower latency than E5-series retrievers, which is consistent with the retrieval time results shown in Fig. 5, confirming that sparse retrieval methods hold a significant efficiency advantage over dense embedding-based approaches.

When the RAG system is subjected to BRRA attacks, all models experience substantial increases in end-to-end response time, with T_{attack} increasing by approximately 15–21 seconds relative to T_{base} across all configurations. This overhead primarily originates from the adversarial document generation process, which involves reward-based iterative optimization and feedback document generation. Particularly noteworthy is that this cost is incurred entirely as an *offline preprocessing cost* on the attacker's side, meaning that end users experience no perceptible increase in query response latency during actual deployment. This characteristic further underscores the covert nature of BRRA attacks, as the attack overhead remains completely transparent to the victim system and its users.

With the dual-stage defense enabled, $T_{defense}$ introduces an additional overhead of 1.5–9.5 seconds beyond T_{attack}, arising from the multi-query perturbation strategy at the retrieval stage and the fairness constraint mechanism at the generation stage. This disparity across models reflects the varying inference costs associated with fairness-guided response regeneration: DeepSeek-R1 exhibits the largest defense overhead due to its slower inference speed, while Llama-3-8B shows the smallest increment.

While the dual-stage defense adds a 1.5–9.5 s latency, this overhead represents a necessary trade-off for security in high-stakes domains like medical RAG, where fairness outweighs marginal delays. For real-time production environments, this latency can be significantly mitigated through engineering optimizations: (1) parallelizing the execution of multi-query perturbations; (2) adaptive triggering, where the full defense is only activated if the initial fairness score falls below a safety threshold; and (3) asynchronous validation, using streaming outputs to mask background fairness alignment. These strategies ensure that robust protection against BRRA attacks remains compatible with the strict latency requirements of real-world deployments.

VIII. CONCLUSION AND LIMITATIONS

A. Limitations

While the proposed BRRA framework demonstrates significant effectiveness across multiple models, datasets, and retrieval configurations, several limitations warrant acknowledgment and motivate future research directions.

The current study focuses on single-round retrieval with representative sparse and dense retrievers, which reflects the dominant RAG deployment paradigm. As the field evolves toward more sophisticated architectures, understanding whether and how bias amplification manifests in these settings becomes increasingly important. In iterative retrieval systems, the multi-round query refinement process may create new opportunities for bias injection at each retrieval step, potentially amplifying bias more aggressively than single-round systems. In hybrid retrieval systems that fuse sparse and dense signals, the interaction between different retrieval modalities may create complex bias propagation patterns not observed in our current experiments. Most importantly, knowledge graph-based RAG systems such as GraphRAG [47] represent a structurally distinct retrieval paradigm where bias may be embedded in entity relationships and graph topology rather than document content, opening an entirely new research dimension for bias amplification analysis.

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We hope our work serves as a foundation for investigating these architectures.

This paper includes a preliminary exploration of defense mechanisms as a first step toward securing RAG systems against bias amplification attacks. As noted in Section VI, the proposed dual-stage defense demonstrates promising results and serves to validate the feasibility of mitigation. However, We emphasize that the proposed defense is intended as a feasibility proof and research baseline rather than a production-ready solution. Our experiments show an attack-defense imbalance. BRRA can still bypass current defenses in some model settings. This means we need better and more principled defenses, such as bias-aware retrieval auditing, adversarial training for embedding models, and certified robustness methods for fairness.

B. Ethical Considerations

This study systematically explores a novel attack framework, BRRA, which amplifies social biases such as gender, age, and disability in RAG systems. We emphasize that the primary motivation for this work is to expose critical security vulnerabilities in current RAG architectures to facilitate the development of more robust, fairness-aware defense mechanisms. By adopting a red-teaming perspective, we demonstrate that even unpoisoned LLMs can be steered toward biased outputs through external knowledge manipulation. We advocate for the responsible deployment of RAG systems, particularly in sensitive domains like healthcare or law, where bias amplification can lead to systemic unfairness. All experiments were conducted in controlled settings using publicly available datasets for academic research purposes only.

C. Conclusion

Previous work has mainly focused on the impact of poisoning attacks on RAG systems regarding the output quality of large language models, such as exacerbating model hallucinations. However, existing studies have overlooked the potential impact of such attacks on model output bias, namely that security vulnerabilities in RAG systems may amplify model bias. Our paper proposes a novel Bias Retrieval Reward Attack framework, which injects carefully crafted adversarial documents into the RAG knowledge base, uses subspace projection techniques to manipulate embedding spaces and influence retrieval results, and constructs a cyclic feedback mechanism during the generation phase to continuously guide large models to generate biased content. Experimental results show that RAG poisoning attacks not only significantly amplify the bias of large models but also severely worsen the unfairness of model outputs, revealing the deep connection between RAG system security and model fairness. To defend against this threat, we further propose a dual-stage defense mechanism that applies protective measures in both the retrieval and generation phases. Experiments validate that this defense method can effectively mitigate attack impacts, providing a feasible solution to build more secure and fair RAG systems. In future work, we will investigate more robust defense mechanisms to mitigate various security threats to RAG systems and ensure the fairness of large models.

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